

Part A · Multiple Choice

2 points each

1. **d) porosity, permeability, and natural recharge by precipitation**

p. 258 — porosity is the pore space between grains; permeability is whether those pores connect; recharge keeps the supply renewing.

2. **a) the water table**

p. 258 — water infiltrates downward through permeable rock until it meets an impermeable layer, which forms the base of the aquifer.

3. **b) About one-third**

p. 259 — it lies 30 to 100 meters deep across about 440,000 square kilometers, and most of its water dates from the last ice age.

4. **b) About eight times as much**

p. 259 — at current rates of use, roughly 70% of the aquifer could be gone by 2050, with serious implications for food production in the central plains.

5. **c) aphotic or profundal zone**

p. 257 — with no sunlight there is no photosynthesis, so most organisms there are scavengers such as crabs and catfish, aided by decomposing fungi and bacteria.

6. **a) saltwater incursion into the aquifer**

p. 259 — seawater enters the depleted aquifer and contaminates it, making the water less useful for drinking and irrigation. It is a recognized problem in developed coastal regions such as Hawaii.

Part B · True or False

1 point each

7. **TRUE**

p. 257 — most were carved by the glaciers that covered Canada during the last Ice Age. Most of the world's lakes are in the Northern Hemisphere for this reason.

8. **FALSE**

Groundwater is renewable, and its use is sustainable as long as the water pumped out is replenished. The problem arises when withdrawal outpaces recharge, as at the Ogallala. (p. 259)

Part C · Short Answer

3 points each

9. Porosity is the amount of open pore space between grains — how much water the rock can hold. Permeability is whether those pores connect — whether the water can travel through. An aquifer needs both: space to store water and connections that let it move to a well.

Rubric: 1 pt porosity defined — the small spaces between sediment grains where water droplets sit. 1 pt permeability defined — the connections between those pores, allowing water to travel through the rock. 1 pt why both are required — a highly porous rock with unconnected pores holds water but cannot yield it, and connections are useless without space to store water in the first place. Credit noting that the impermeable layer beneath is what forms the base of the aquifer. (p. 258)

10. The water table falls, forcing wells to be dug deeper or leaving them dry; the ground surface may subside beneath houses and structures; and in coastal areas seawater may enter and contaminate the aquifer.

Rubric: 1.5 pts each for any two. **Falling water table** — wells must be dug deeper at greater cost and energy, and shallower wells go completely dry. **Subsidence** — lowering the water table can cause the ground surface to sink, including beneath houses and other structures. **Saltwater incursion** — in coastal regions seawater enters the depleted aquifer and contaminates it for drinking and irrigation. (p. 259)